

MODULE DESCRIPTION

Module title	Finance 2 (Investment Finance)
Module code	BFIN
Study program	Finance and Accounting (BFA), Business Administration (BBA), Economics (BSE)

Module coordinator/Lecturer

Type	Lecturer	Email	Office	Office hours
Coordinator	Dr. Le Minh Hanh	hanh.lm@vgu.edu.vn	Cluster1 –R.506	9:00am – 4:00pm
Lecturer	Dr. Omer Cayirli	omer_cayirli@uncbusines.net	Cluster1 –R.522	9:00am – 4:00pm
RTA	Ty Dam Phuong Ngoc	ty.dpn@vgu.edu.vn	Cluster1 –R.531	9:00am – 4:00pm
Tutor				
Other	None			

Classification ☒ Compulsory ☐ Semi-elective

Semester Winter Semester 2026

Student workload

Credits	6	ECTS
Teaching periods	45	AHs
Assignments and independent learning	135	AHs
Total Working hours	180	AHs

Frequency and Eligibility criteria

B-Modules (except BWIA) To be eligible for the B-Modules, students must pass a minimum of 45 credit points from O-modules including OMAT, OSTA

Course objectives

Contents: The study of financial instruments and markets, especially stocks, bonds and derivatives, mostly from an investor perspective.

- Stock investments: i) derivation of an optimal investment portfolio, ii) determination of the cost of capital of firms, and iii) firm valuation. Modern portfolio theory as the basis for determining the cost of equity capital based on the Capital Asset Pricing Model (CAPM) or multifactor models. In addition, explanation of basic approaches to firm valuation.
- Bond investments: Analysis of the yield curve, basic valuation of (risk-free) fixed-income securities (bonds), concepts of duration as well as convexity as key measures of the interest rate sensitivity of bonds, and their deployment evaluation in the context of interest rate risk management.
- Derivatives: Presentation of the basic derivative financial instruments forwards, futures and options and the functioning of the markets for these products. Development of the theoretical models for the determination of the forward/future price as well as options under no-arbitrage conditions based on an analysis of the product characteristics.
- Investigate basic risk management strategies for securities portfolios using derivative financial instruments.

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Competency Objectives: Students...

- ... acquire basic professional and methodological skills in the field of financial economics. This includes, among other things, the ability to explain topics and terms such as optimal investment, asset allocation, players on financial markets, market efficiency, company valuation, performance comparison.
- ... can name and explain the characteristics and modes of operation of the most important financial instruments and the respective financial markets. These are, for example, stocks, bonds, funds, ETFs, forwards, derivatives.
- ... can present the most important models for the analysis and valuation of financial instruments, their assumptions as well as their limitations and apply the models to practical examples. The models include CAPM, APT, binomial model and Black-Scholes.
- ... can present, interpret and apply simple risk management strategies for securities portfolios using derivatives. These are for example PF optimization, duration matching, hedging of existing portfolios and hedging of intended portfolio acquisition.
- ... develop competencies for the independent solution of application-oriented problems from the above-mentioned areas within the framework of the exercise sessions.
- ... are able to explain the learned contents in the context of practical application cases.

Course learning outcomes. The objective of this course is to introduce the theory and empirical methods relevant to investment management. Building on the concepts covered in OFIN, it will discuss the characteristics of various financial securities along with the risks and rewards associated with them. Topics covered include optimal portfolio selection, asset pricing models, market efficiency, investor behavior, and the main derivative markets and instruments.

Course Learning Outcomes (CLO)	Related Program Learning Outcome (PLO)	Level of Contribution (x: <i>Beginner</i> - xx: <i>Intermediate</i> - xxx: <i>Advanced</i>)
Basic professional and methodological skills in the field of financial economics.	BBA (1, 2, 3, 4); BFA (1, 2, 3, 4)	BBA (x, xx, x, x); BFA (x, xx, xxx, x)
Understanding of the characteristics and modes of operation of the most important financial instruments and the respective financial markets.	BBA (1, 2, 3); BFA (1, 2, 3)	BBA (x, xx, x); BFA (x, xx, xxx)
Present the most important models for the analysis and valuation of financial instruments, along with their assumptions and limitations; apply the models to practical examples.	BBA (1, 2, 3); BFA (1, 2, 3)	BBA (x, xx, x); BFA (x, xx, xxx)
Present, interpret and apply simple risk management strategies for securities portfolios using derivatives.	BBA (1, 2, 3, 4, 5); BFA (1, 2, 3, 4, 5)	BBA (x, xx, x, x, x); BFA (x, xx, xxx, x, x)
Develop competencies for the independent solution of application-oriented problems related to asset valuation and management.	BBA (1, 2, 3, 4, 5); BFA (1, 2, 3, 4, 5)	BBA (x, xx, x, x, x); BFA (x, xx, xxx, x, x)
Explain and use the learned contents in the context of practical application cases.	BBA (1, 2, 3, 4, 5); BFA (1, 2, 3, 4, 5)	BBA (x, xx, x, x, x); BFA (x, xx, xxx, x, x)

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Program's learning outcomes:

ILO1: Theoretical Knowledge of Economics and Business Science (general and field specific)

- Identify and discuss economics and business issues based on the reflection against fundamental concepts and theories;
- Integrate knowledge from various subjects to develop and evaluate comprehensive business solutions on a scientific basis.

ILO2: Quantitative methods and critical thinking skills (general and cross-disciplinary)

- Apply quantitative methods and statistical tools to soundly analyze and evaluate economic and business data based on profound theories and frameworks to develop research questions and hypotheses in the field and solve these questions with sustainability and responsibility.

ILO3: Knowledge (field specific)

- Apply specific concepts, theories and methods of their specialization (in Management and Business Informatics for BBA students and in Finance and Accounting for BFA students) to analyze and develop business solutions.

ILO4: Ethical reasoning and sustainability mindset (general and cross-disciplinary)

- Integrate ethical reasoning and a sustainability mindset to evaluate and develop business and economic decisions.

ILO5: Communication skills and global competences (general and cross-disciplinary)

- Effectively use cross-cultural communication skills for academic and professional purposes

Contents (Outline of subject and topics explored)

No.	Topic	Teaching hours	Assignments and independent learning hours
1.	Introduction; The Investment Environment; Asset Classes and Financial Instruments	3	9
2.	Fixed-Income Securities	8	24
3.	Portfolio Theory and Practice	8	24
4.	Asset Pricing Models	7	21
5.	Market Efficiency, Behavioral Finance and Market Anomalies	3	9
6.	Futures, Forwards and Swaps	6	18
7.	Option Markets & Option Valuation	10	30

Learning activities. This course will utilize lectures, exercises, case studies, and self-study to attain the learning goals. Python, VS Code/Antigravity and Jupyter Notebooks will be used to enhance learning activities. The lectures will adopt an interactive approach; students should spend sufficient time on assigned readings/tasks for each lecture before the class. The course RTA will deliver 6 (six) recitation sessions (12 ATHs, optional) to support learning progress.

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Activity	Description	Alignment with ILOs	Estimated Student Effort (Hours)
Lectures	Theoretical fundamentals and practical applications	BBA (1, 2, 3, 4, 5); BFA (1, 2, 3, 4, 5)	30
Tutorials/Exercises	Utilization of theoretical knowledge	BBA (2); BFA (2)	5
Case Studies	Transfer the theoretical knowledge to complex real-world scenarios	BBA (2, 3, 4, 5); BFA (2, 3, 4, 5)	10
Assignment	Enhance theoretical knowledge and transfer the acquired skills to practice	BBA (1, 2, 3, 4, 5); BFA (1, 2, 3, 4, 5)	10
Self-study	Preparation for lectures, internalization of concepts	BBA (1, 2, 3, 4, 5); BFA (1, 2, 3, 4, 5)	125
Recitation Sessions	Content review, enhancing learning progress	BBA (1, 2, 3); BFA (1, 2, 3)	12 (Optional)

Mode of assessment

The mode of assessment for this course is based on the portfolio approach. The portfolio consists of 2 (two) group assignments, and a final exam (75 minutes). Participation and contribution to the in-class discussions are important parts of the portfolio. No weights are assigned to individual parts of the portfolio. The final grade for this course will be based on the students' progress and overall performance.

Regular exam - Pass requirements:

A student is required to pass all assessment items to pass the course (competency and/or grade-based): **NO**
 A student is required to attempt all assessment items to pass the course: **YES**

Retake exam - Pass requirements:

The portfolio approach will be applicable for the course grade in case a retake exam is needed. Except class contribution/participation component, portfolio elements from the regular examination cannot be carried forward to the retake examination. The pass requirements for the retake exam are the same as the regular exam (see above).

Module materials

Required texts:

- 1) Investments, 12th edition, 2021. Zvi Bodie, Alex Kane, Alan J. Marcus. McGraw-Hill.

Recommended texts:

- 1) Corporate Finance, Global Edition Sixth Edition, Jonathan Berk and Peter DeMarzo, 2024. Pearson.
- 2) Modern Portfolio Theory and Investment Analysis, 9th edition, 2014. Edwin J. Elton, Martin J. Gruber, Stephen J. Brown, William N. Goetzmann. Wiley.

Lecture Notebooks: Lecture notebooks will be available for access on GitHub before each class (without solutions to end of lecture exercises) and will be updated after each class (with solutions). The supporting lecture materials will be uploaded to Moodle as lectures progress.

Written/updated by
 Dr. Omer CAYIRLI
 Date 27 / 07 / 2026

Approved by

 Date / /2026

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DETAILED COURSE SCHEDULE

Lecture 1 - Introduction

- General introduction to the course, a brief overview of contents, and learning goals.
- The Investment Environment; Asset Classes and Financial Instruments
- **Reading(s): BKM Ch. 1 & 2**

Lecture 2 – Fixed-Income Securities I

- Bond Prices and Yields
- The Term Structure of Interest Rates
- **Reading(s): BKM Ch. 14 & 15**

Lecture 3 – Fixed-Income Securities II

- Managing Bond Portfolios
 - ✓ Interest Rate Risk & Convexity
 - ✓ Passive Bond Management
- **Reading(s): BKM Ch. 16**

Lecture 4 – Portfolio Theory and Practice I

- Risk and Return
- Capital Allocation to Risky Assets
- **Reading(s): BKM Ch. 5 & 6**

Lecture 5 – Portfolio Theory and Practice II

- Efficient Diversification
- **Reading(s): BKM Ch. 7**
- **Group Assignment I**

Lecture 6 – Index Models and The Capital Asset Pricing Model

- The Single-Index Model
- The Capital Asset Pricing Model
- **Reading(s): BKM Ch. 8 & 9**

Lecture 7 – APT and Multifactor Models of Risk and Return

- Multifactor Models
- Arbitrage Pricing Theory
- **Reading(s): BKM Ch. 10**

Lecture 8 – The Efficient Market Hypothesis & Behavioral Finance

- The Efficient Market Hypothesis
- The Behavioral Critique
- Empirical Evidence on Security Returns
- **Reading(s): BKM Ch. 11, 12.1 & 13**

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Lecture 9 –Derivatives I

- Introduction to Derivatives
- Futures and Forwards
- **Reading(s): BKM Ch. 22 Lecture 10 – Derivatives II**

Lecture 10 – Derivatives II

- Futures, Swaps, and Risk Management
- **Reading(s): BKM Ch.23**

Lecture 11 – Derivatives III

- Option Markets
- **Reading(s): BKM Ch. 20**

Lecture 12 – Derivatives IV

- Option Valuation
- **Reading(s): BKM Ch. 21**